

Bocconi

COMPUTER SCIENCE

code 30424 a.y. 2024-2025

Lesson 1

Understanding the ICT



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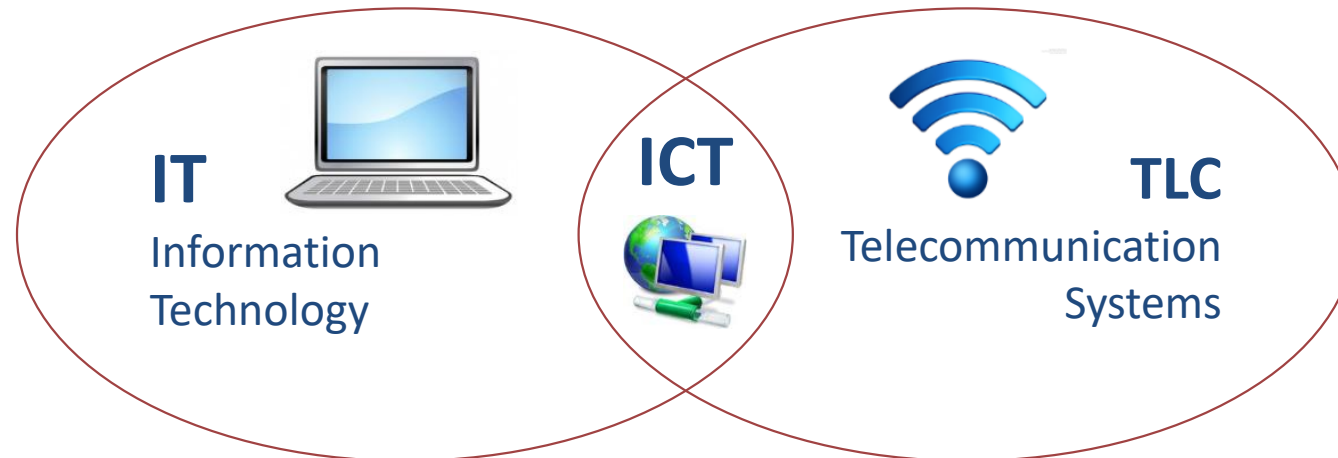
Objectives of the lesson

- Understand what the term ICT refers to
- Understand the evolution of ICT
- Understand the development prospects of ICT tools

Defining ICT

ICT = Information and Communication Technology

It is the set of all technologies that allows to store, process and analyze **data** to produce and communicate **information** through digital tools



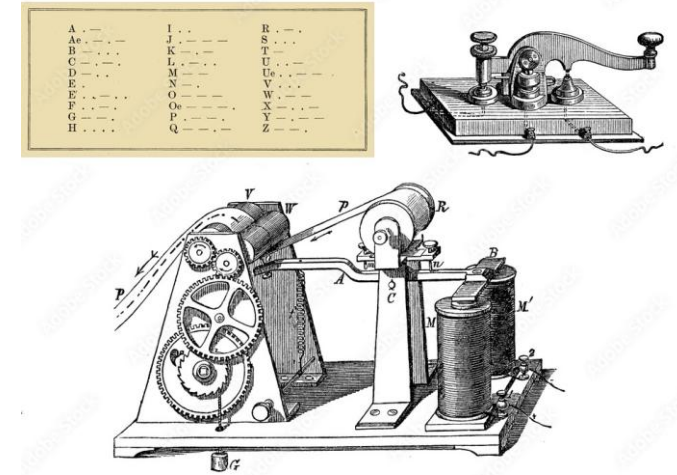
It is the integration between information technology (IT) and telecommunication systems (TLC)

The evolution of telecommunication systems



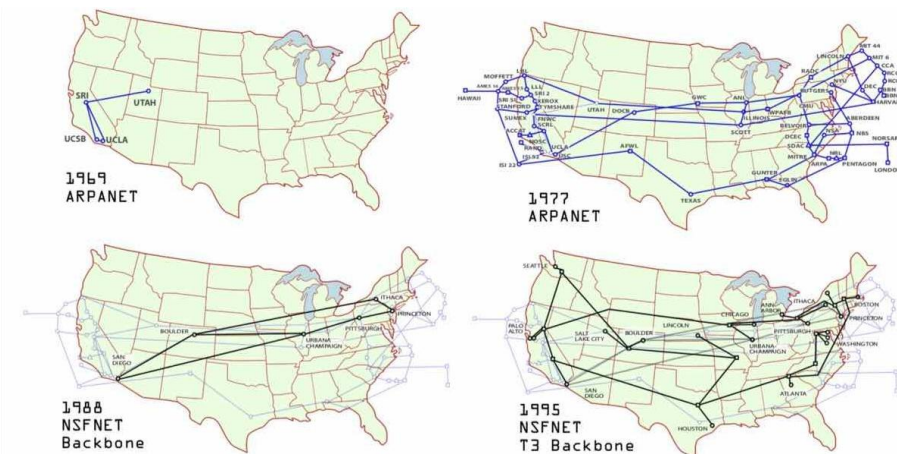
Telecommunications: from wired to wireless

- **Telegraph (1837):**
Samuel Morse's invention introduced long-distance wired communication using Morse code
- **Telephone (1876):**
Alexander Graham Bell's innovation enabled real-time voice communication
- **Radio Transmission (1900s):**
Guglielmo Marconi introduced the first wireless communication system in 1896. Further developments led to the evolution from point-to-point communications to broadcasting



Telecommunications: from local to global

- **Satellite Communications (1960s):**
Enabled global communication through space-based relays
- **ARPANET (1969):**
Developed for research and defense networks, it forms the foundation of the modern Internet
- **TCP/IP Protocol (1983):**
Standardized communication protocols, which have made the Internet globally interoperable
- **World Wide Web (1991):**
Tim Berners-Lee introduced the Web, revolutionizing access to information



Telecommunications: from stationary to mobile

- 1G (First Generation - 1980s)
 - Analog voice communication
 - Basic voice calls, low quality and poor security
- 2G (Second Generation - 1991)
 - Digital communication (GSM, CDMA)
 - Introduced SMS, voice encryption and better call quality
- 3G (Third Generation - 2001)
 - Mobile broadband (UMTS, HSPA)
 - Internet access, video streaming and speeds up to 2 Mbps



Telecommunications: from stationary to mobile

- **4G** (Fourth Generation - 2009)
 - LTE technology
 - High-speed internet (up to 1 Gbps), HD streaming, and VoIP
- **5G** (Fifth Generation - 2019)
 - Ultra-fast speeds (up to 10 Gbps) and low latency (1 ms)
 - Real-time applications (AR/VR, IoT, autonomous vehicles)
- **6G** (Sixth Generation - ~2030)
 - Expected speeds of 1 Tbps and sub-millisecond latency
 - Integration of AI and brain-computer interfaces



The evolution of information technologies

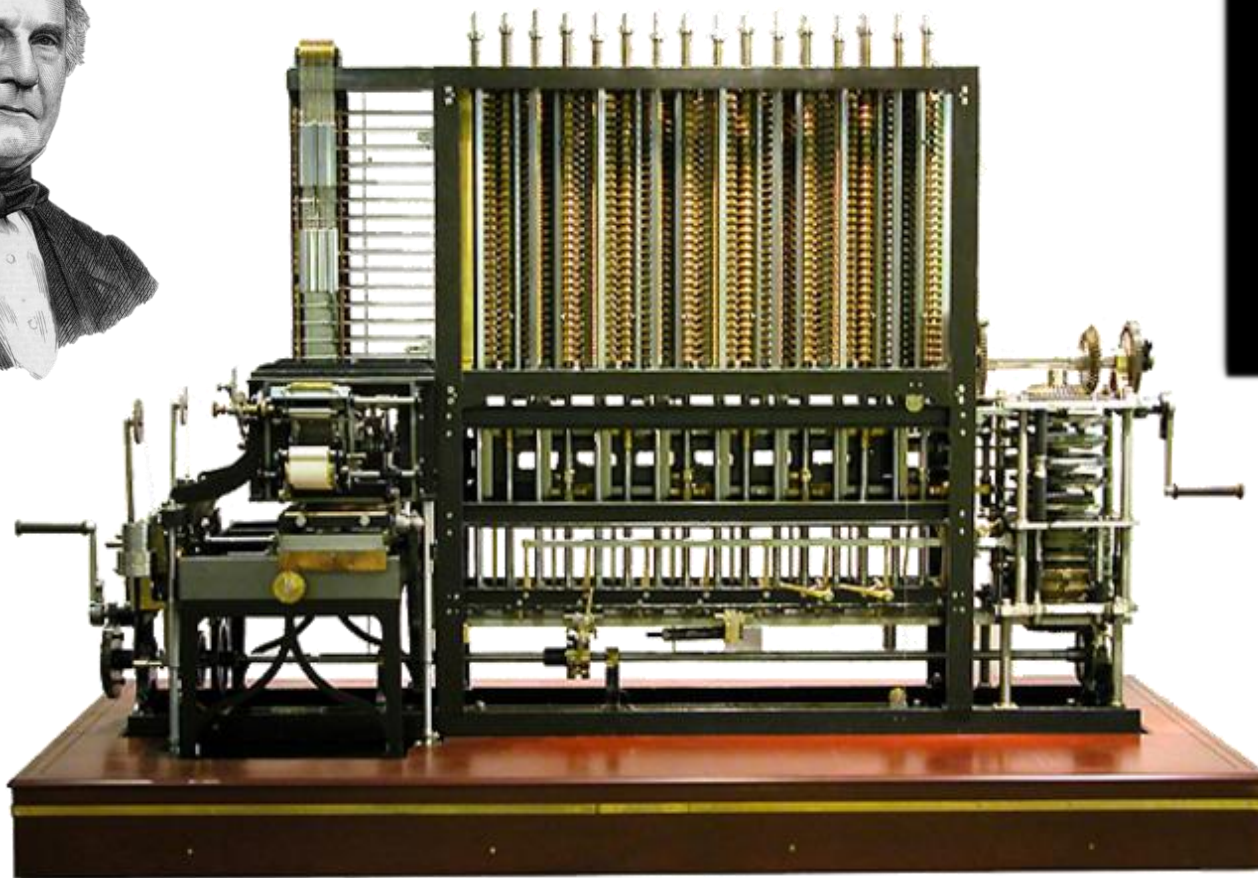
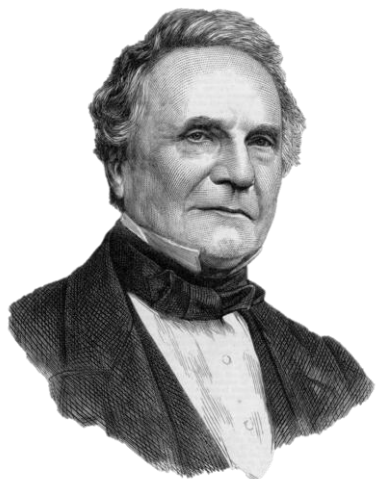


Information technologies from 1600 to 1800

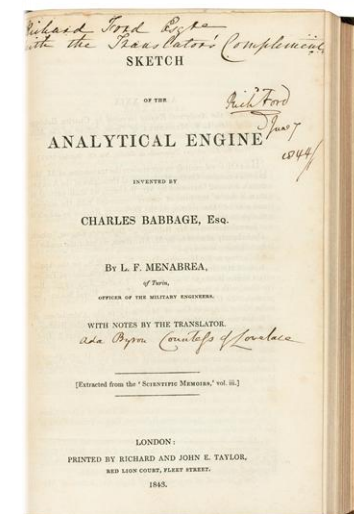
- **Blaise Pascal (1642)**
 - Invented the Pascaline, a mechanical calculator capable of performing addition and subtraction
 - Early example of mechanized computation
- **Charles Babbage (1830s)**
 - Designed (but not physically assembled) the Analytical Engine, considered the first concept of a programmable mechanical computer
 - Introduced key principles of computation, like control flow and memory
- **Ada Lovelace (1840s)**
 - Worked with Babbage and wrote the first algorithm designed for the Analytical Engine
 - Recognized as the first computer programmer



Information technologies from 1600 to 1800



The Babbage Engine
at the Computer History Museum
Mountain View, California

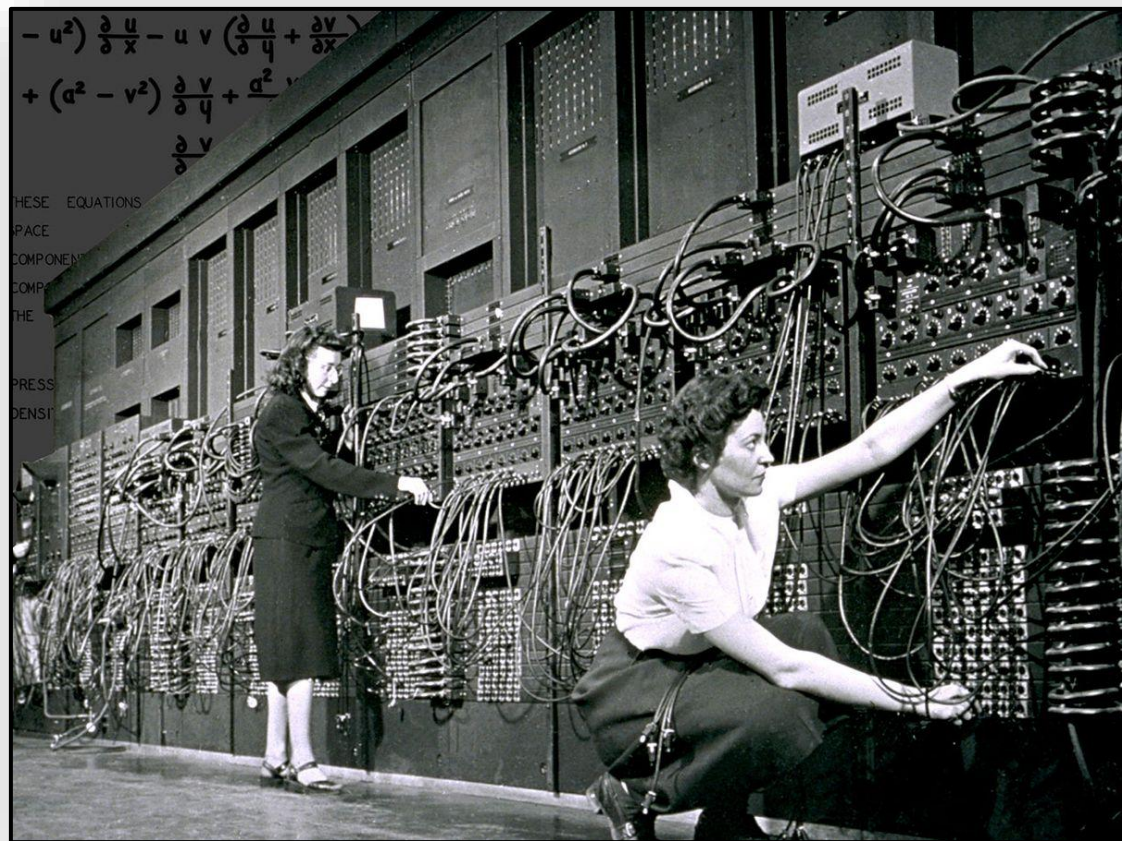


Information technologies in first half of 1900

- **Alan Turing (1930s-1940s)**
 - Developed the concept of the Turing Machine, a theoretical model that demonstrated how any computation could be performed using a series of instructions (algorithm)
 - Considered one of the fathers of artificial intelligence and modern computing
- **John von Neumann (1945)**
 - Proposed the Von Neumann Architecture, the basis of most modern computers
 - Introduced the idea of a stored-program computer, where both data and instructions are stored in the same memory
- **Early Computers (1940s-1950s)**
 - ENIAC (1945): The first general-purpose electronic computer using vacuum tubes
 - UNIVAC (1951): First commercial computer designed for business and government use



Early Computers



Marlyn Wescoff [left] and Ruth Lichterman: two of the programmers of ENIAC
(PHOTO: CORBIS/GETTY IMAGES)

The ENIAC

<https://www.youtube.com/watch?v=bGk9W65vXNA>

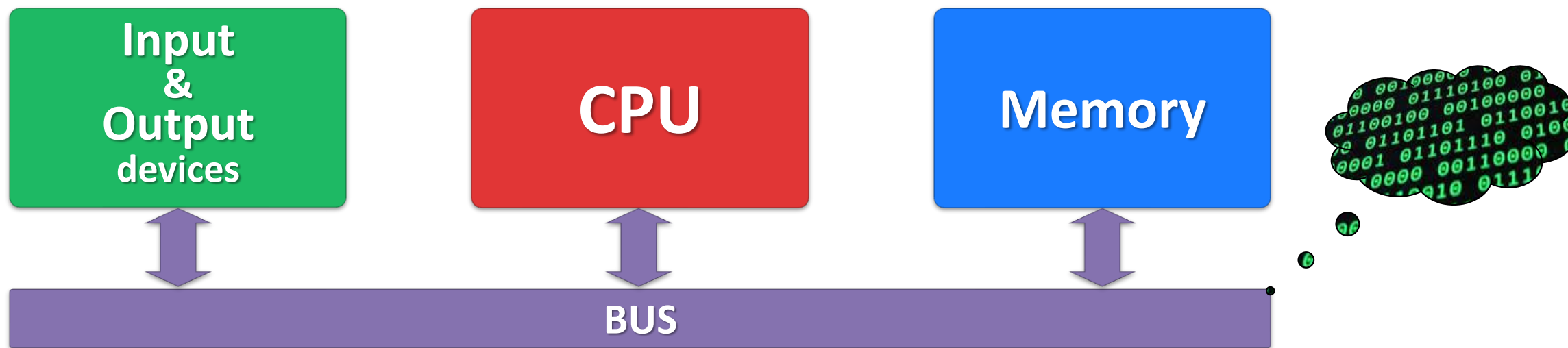
The first programmers

<https://vimeo.com/ondemand/eniac6/155075248>

(Interviews dates back to 1997)



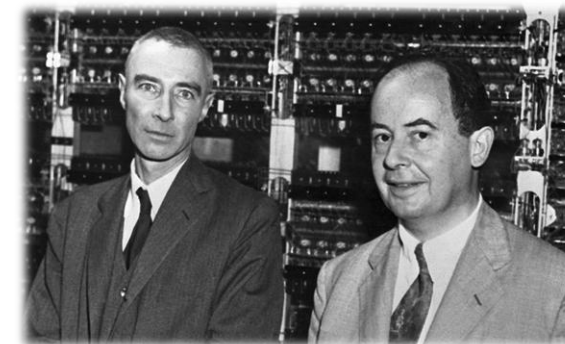
Von Neumann Computer Architecture



All these components rely on **bistable devices** that can handle just two possible status:

- The voltage level of an electrical conductor: **high** or **low**
- The state of magnetic polarization of a surface: **positive** or **negative**
- The state of the light in an optical fiber: **on** or **off**

Data and instructions are all coded by elements in **Binary Notation**, conventionally represented with strings of "0" and "1"



Information technologies in second half of 1900

- **Development of Programming Languages (1950s-1960s)**
 - Creation of early programming languages like FORTRAN (1957 - for scientific computing), COBOL (1959 - for business applications), and Assembly Language
 - Rise of software development started to make computers more accessible
- **Personal Computers and Microprocessors (1970s-1980s)**
 - Invention of the microprocessor (Intel 4004 in 1971) to replace vacuum tubes
 - Rise of personal computers started with the Apple II (1977) and IBM PC (1981)
- **Internet and World Wide Web (1980s-1990s)**
 - Adoption of TCP/IP protocols (1983) and introduction of URLs, HTTP and HTML
 - The World Wide Web (1991) revolutionized information access and sharing



Information technologies in the new century

- **Mobile and Social Revolution (2000s-Present)**
 - Emergence of smartphones, mobile apps, and cloud computing
 - Social Media platforms redefined global interaction
- **IoT and Cloud services (2010s-Present)**
 - Integration of computing into everyday devices through the 'Internet of Things'
 - Cloud-native applications and platforms became standard for businesses
- **AI, VR and Quantum Computing (2020s-Present)**
 - Rapid advances in AI, machine learning, and quantum computing
 - Development of immersive digital experiences through augmented reality (AR) and virtual reality (VR) applications



What about today?



The world fastest supercomputers

For more videos:

[El Capitan \[1\]](#)

[El Capitan \[2\]](#)

...And what about tomorrow?



Artificial Intelligence

For more videos:

[AI \[1\]](#)

[AI \[2\]](#)



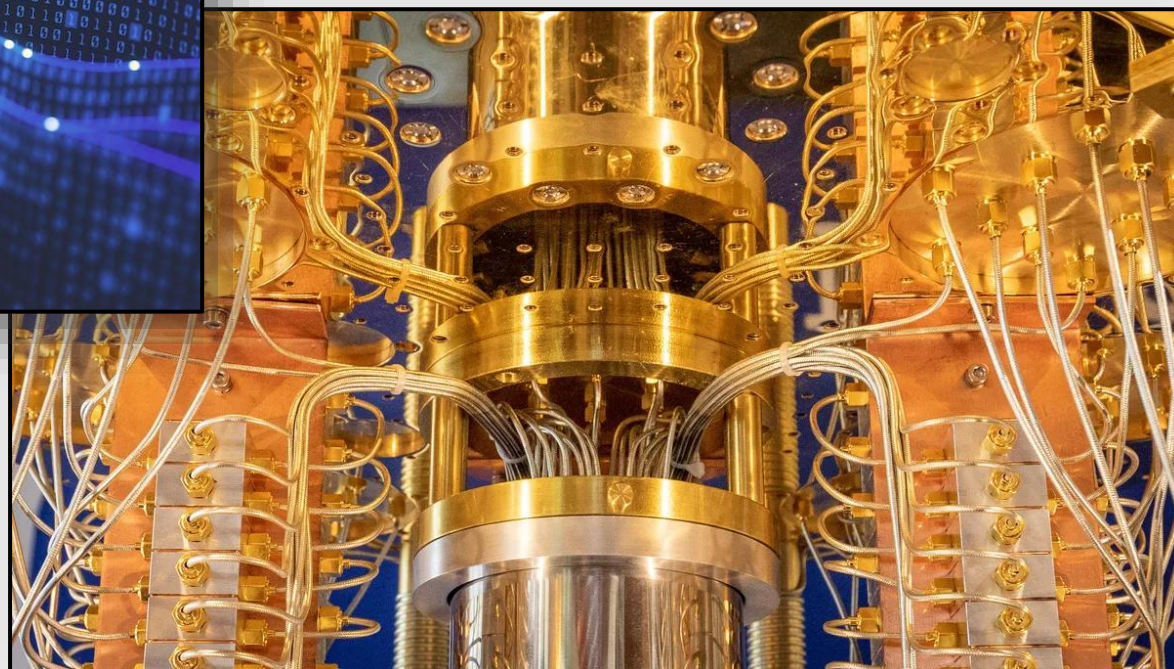
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Quantum computers

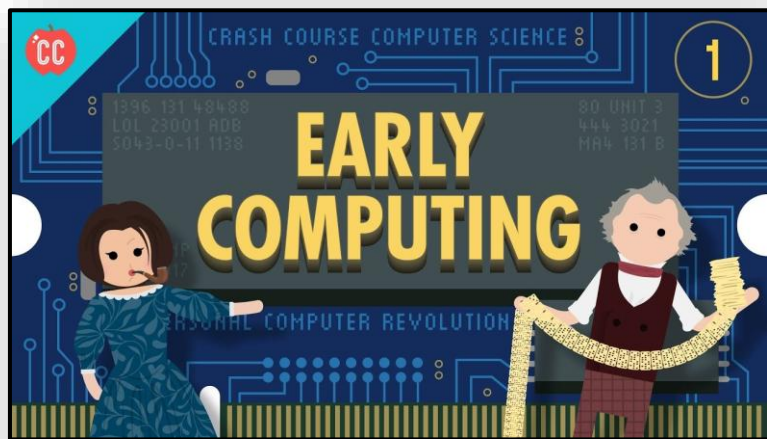
For more videos:

[QC \[1\]](#)

[QC \[2\]](#)

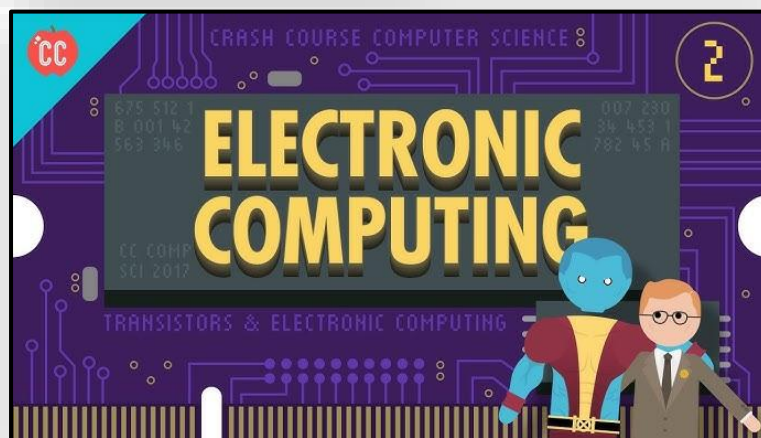


To know more about our history...



CHM Computer
History
Museum

Timeline of Computer History



Evolution of Technologies
in the last decades

What should we know about ICT?



ICT tools

Three macro categories:

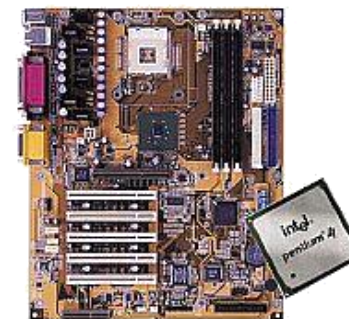
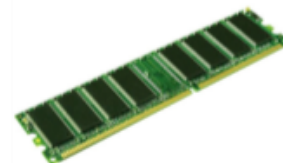
- Hardware
- Software
- Networks



Hardware: basic concepts

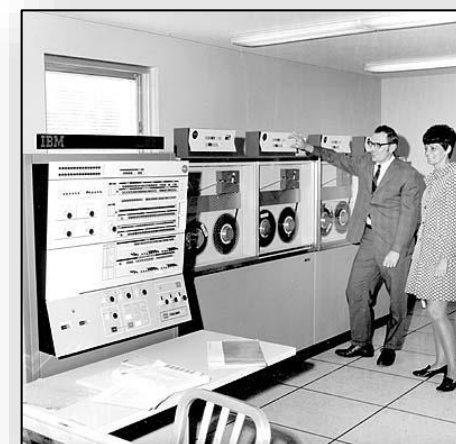
In order to work effectively using ICT tools it is important:

- to know the architecture of a computer
- to learn what they are and what they do: Motherboard, CPU, RAM, ROM, ...
- to learn the different storage devices and their characteristics
- to learn the input and output devices and the different types of connection to the PC
- to learn which factors influence the performance of a PC



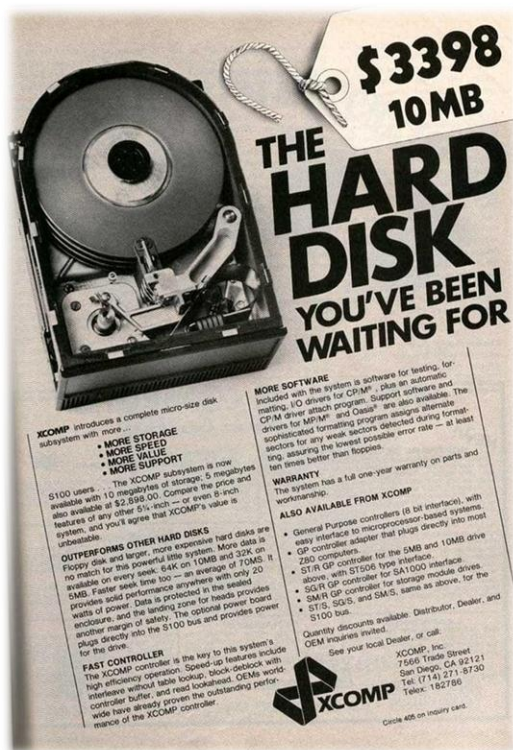
Hardware: main trends

- Increased processing power
- Increased memory capacity
- Reduced size
- Extended functionality



Hardware: Increased memory capacity

1984



1 MB = \$ 339.8

2024 (consumer)

Model Number: WD20EARZ

WD Blue PC Desktop Hard Drive - 2TB

★★★★★ 3.9 (96) Write a review

WD Blue internal hard drives deliver reliability for office and web applications. They are ideal for use as primary drives in desktop PCs and for offi ... [See more](#)

Disk Speed

7200 RPM

5400 RPM

Capacity

2TB

\$59.99



1 MB = \$ 0.000028

2024 (enterprise)

Model Number: WD241KRYZ

WD Gold Enterprise Class SATA HDD - 24TB

★★★★★ 4.0 (114) Write a review

Customize your business's storage mix to fit your specific needs with a full portfolio of enterprise-class WD Gold HDDs in capacities of 1TB to 24TB¹, ... [See more](#)

Capacity

24TB

Cache Size

512MB

\$529.99 ~~\$599.99~~



1 MB = \$ 0.000021



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Software: basic concepts

In order to work effectively using ICT tools it is important:

- to know what is an operating system
- to learn using correctly at least one operating system
- to know what is an application program
- to know what application programs we need



Software: main trends

- Ease of use
- Ubiquitous use (MobileApp)
- New access modalities (WebApp, SaaS)
- Evolution of programming languages



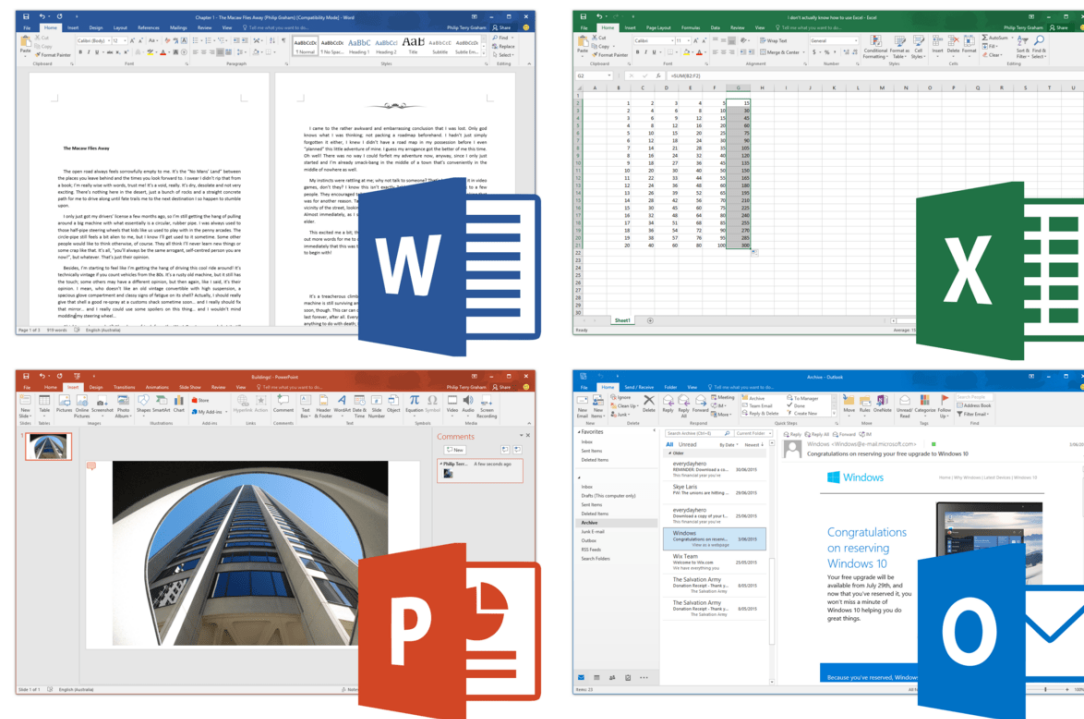
Programming languages

A programming language is a formal language that enables to provide a computer with the instructions needed to solve a problem



Application software

Application software consists of programs used directly by users to process data and produce the information they need



Examples of application software

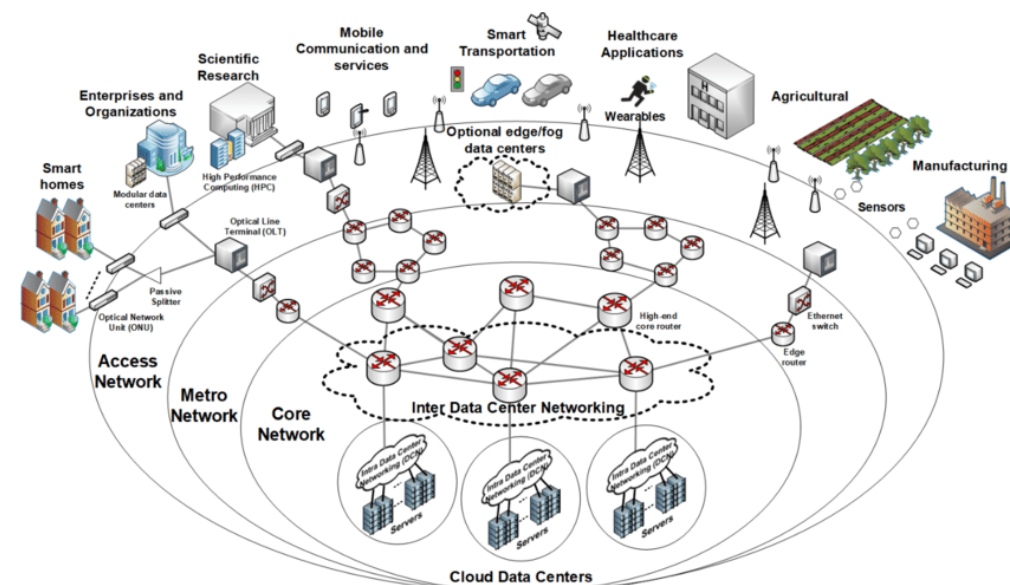
Application	Use	Examples
Worksheets	Managing spreadsheets based on cell tables where formulas can be inserted, graphic processing of data	MS Excel, OpenOffice.org Calc
Word processing	Word processing for creating documents and formatting texts	MS Word, OpenOffice.org Writer
Presentation tools	Creating multimedia presentations with slides which can be viewed in sequence	MS PowerPoint, OpenOffice.org Impress
Data Base Management Systems	Filing, structuring and querying data bases	MS Access, OpenOffice.org Base
Software for statistical processing	Controlling and analysing large quantities of data, econometric analysis	SPSS, SAS, Stata, R, EViews
Software for mathematical processing	Studies of functions, graphic representations, ...	Matlab, Mathcad
Graphic applications	Photo editing, CAD, ...	Adobe Photoshop, AutoCAD
Utilities	Antivirus, PDF management, ...	McAfee VirusScan, Adobe Reader
Web applications	Individual productivity software which can be used directly via Web through the browser	Google Docs, webmail, search engines, ...



Networks: basic concepts

In order to work effectively using ICT tools it is important:

- to understand the benefits of a network infrastructure
- to know the taxonomy of networks based on their type (wired/wireless), extension (LAN/WAN/WLAN) and architecture (Client/Server, Peer to peer)
- to know what the Internet is and what the World Wide Web is
- to know what TCP/IP and HTTPS protocols are



Networks: main trends

- Ultra-broadband, optical fiber and 5G connectivity
- Cloud computing
- Internet of Things
- From Web 2.0 to Web 3



Networks: Cloud computing

- It is an increasingly widespread way of using resources available on the Internet
- It consists of a set of technologies that allow to use remote, virtualized and distributed resources through the Internet:
 - Hardware (CPU, storage) -> IaaS (Infrastructure as a Service)
 - Software (OS, applications) -> SaaS (Software as a Service)
 - Development platforms -> PaaS (Platform as a Service)



Networks: Ultra-broadband

- The spread of broadband is considered a key factor in the economic growth of a country
- The connection speed is an essential technical requirement for most part of Cloud services but also for personal and business productivity
- There are other parameters than the connection speed that are crucial for an efficient connection
- Several services allow to measure the most relevant parameters of a connection (e.g. www.speedtest.net)



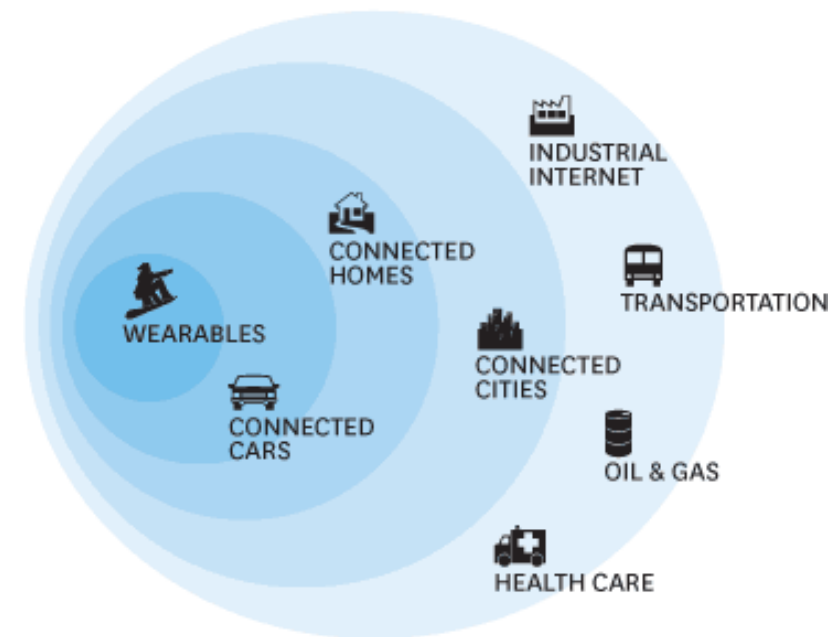
The Digital Divide

- The term Digital Divide refers to the gap between those who can access new technologies and those who do not have access to them
- The causes of the Digital Divide are mainly of two types:
 - Limited access to adequate ICT infrastructures
 - Computer illiteracy of users



The Internet of Things

- The Internet of Things (IOT) is the part of the Internet that concerns the interconnections between objects (smart devices)
- The fields of application are many:
 - Home automation
 - Automotive
 - Smart cities
 - Industrial automation
 - Electronic payments
 - ...



Knowledge to acquire with today's lecture

- Definition of ICT
- Evolution of telecommunication systems
- Evolution of information technologies
- Von Neumann computer architecture
- Components of the three ICT categories
- Main features and trends of ICT tools

Study materials

This **set of slide** has to be considered the study material for contents of **Lesson 1**

Any links to external resources have to be considered as supplementary material for further exploration of the topics covered